

Benchmarking Class Activation Map Methods for Explainable Brain Hemorrhage Classification on Hemorica Dataset

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Abstract—Explainable Artificial Intelligence (XAI) has become an essential component of medical imaging research, aiming to increase transparency and clinical trust in deep learning models. This study investigates brain hemorrhage diagnosis with a focus on explainability through Class Activation Mapping (CAM) techniques. A pipeline was developed to extract pixel-level segmentation and detection annotations from classification models using nine state-of-the-art CAM algorithms, applied across multiple network stages, and quantitatively evaluated on the Hemorica dataset, which uniquely provides both slice-level labels and high-quality segmentation masks. Metrics including Dice, IoU, and pixel-wise overlap were employed to benchmark CAM variants. Results show that the strongest localization performance occurred at stage five of EfficientNetV2-S, with HiResCAM yielding the highest bounding-box alignment and AblationCAM achieving the best pixel-level Dice (≈ 0.57) and IoU (≈ 0.40), representing strong accuracy given that models were trained solely for classification without segmentation supervision. To the best of current knowledge, this is among the first works to quantitatively compare CAM methods for brain hemorrhage detection, establishing a reproducible benchmark and underscoring the potential of XAI-driven pipelines for clinically meaningful AI-assisted diagnosis.

Index Terms—Explainable Artificial Intelligence (XAI), Class Activation Map (CAM), Brain Hemorrhage Classification, Deep Learning

I. INTRODUCTION

Intracranial hemorrhage (ICH) is a life-threatening condition that requires rapid and accurate diagnosis, as delays in treatment can significantly increase morbidity and mortality [1]. Non-contrast head CT is the standard imaging modality for detecting hemorrhages; however, interpretation is often challenging due to subtle intensity differences, variability in lesion shape and size, and the presence of confounding pathologies [2]. Deep learning models have demonstrated strong performance in slice-wise hemorrhage detection, but their “black-box” nature limits clinical adoption. In high-stakes scenarios such as ICH diagnosis, automated algorithms alone are insufficient. Clinicians require transparent explanations to

validate predictions and maintain trust in the system. This has motivated a growing body of research in explainable AI (XAI), where new methods for interpretable deep learning are introduced at an accelerating pace [3].

Progress on XAI for brain hemorrhage has been constrained by the lack of high-quality pixel-level annotations [4]. The recently released Hemorica dataset [5] helps to address this limitation. Unlike widely used datasets such as RSNA [6] and CQ500 [7], which lack detailed segmentation masks, Hemorica contains 327 head CT studies with slice-level labels and pixel-wise annotations across multiple hemorrhage subtypes. This makes it uniquely suited not only for evaluating classification models but also for quantitatively benchmarking XAI methods against ground-truth segmentations.

The primary focus of this study is a systematic comparison of state-of-the-art Class Activation Mapping (CAM) [8] techniques for visual explanation in the context of ICH. While XAI in medical imaging includes both textual [9] and visual [10] approaches, visual explanations, particularly CAM-based methods, are most compatible with radiology workflows since they highlight the image regions most responsible for a model’s decision. Previous evaluations of CAMs for ICH have been limited by small or non-diverse datasets and the absence of reliable segmentation masks [11]. Leveraging the detailed annotations in Hemorica, this study provides a comprehensive quantitative and qualitative comparison of multiple CAM variants across different network stages, enabling assessment of which methods produce clinically meaningful and spatially coherent explanations.

In parallel with the XAI analysis, the classification task itself is revisited. To ensure that CAM comparisons are conducted under strong baselines, classification performance is optimized through controlled experiments that examine the effects of input resolution, augmentation strategies, class imbalance handling, decision thresholds, and backbone architectures.

The contributions of this study are summarized as follows:

- 1) Baseline results on the Hemorica classification task are revisited. An in-depth analysis of the deep-learning

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models used in prior work is provided, and training and evaluation procedures are refined to obtain improved classification results.

- 2) A comparative analysis of state-of-the-art CAMs is presented. These methods are evaluated both qualitatively and quantitatively on the brain hemorrhage classification task, and consistent evaluation metrics are reported to facilitate reproducibility.

The remainder of this paper is structured as follows. Section II reviews prior research on brain hemorrhage diagnosis and explainable AI approaches. Section III details the proposed methodology, covering the Hemorica dataset, the CAM variants under study, the EfficientNetV2 backbone, and the evaluation metrics. Section IV presents the experimental setup and evaluation protocol. Section V reports classification performance and the comparative analysis of CAM-based localization. Finally, Section VI concludes the study and outlines possible directions for future research.

II. RELATED WORKS

Deep learning has been increasingly applied in medical imaging [12], [13], including brain hemorrhage diagnosis [14], [15]. A parallel line of research has emphasized providing explanations for model predictions [16]. This section reviews prior studies on brain hemorrhage and explainable AI (XAI) methods relevant to our work.

A. Datasets

Intracranial hemorrhage (ICH) datasets differ in size, annotation quality, and task suitability. The RSNA dataset is one of the largest, with over 25,000 head CT scans labeled at the slice level [6]. However, the absence of pixel-level segmentation restricts its use to classification. The PhysioNet ICH dataset [17] includes segmentation masks, but annotation quality is coarse and the dataset is relatively small. CQ500 provides around 500 high-quality CT studies with expert slice-level labels [7], but no segmentation masks. Seg-CQ500 [18] adds masks for 51 studies, enabling preliminary localization but limited for large-scale evaluation. Other datasets such as INSTANCE [19] and institutional cohorts exist but are not broadly accessible. In contrast, the Hemorica dataset provides 327 non-contrast CTs with slice-level labels and pixel-level segmentation masks across five hemorrhage subtypes, enabling both robust classification benchmarks and quantitative explainability evaluations using CAM-based methods.

B. Brain Hemorrhage Diagnosis

Brain hemorrhage detection has been approached with diverse deep learning models [20], from early CNNs [21] to recent transformer-based architectures [11]. Some studies focus on 2D slice-wise classification [14], while others employ 3D volumetric models to capture inter-slice context [22]. Techniques include ensemble learning [23], multi-view CNNs [24], and attention-based networks [25], often trained on RSNA, CQ500, or PhysioNet datasets.

The original Hemorica study established baseline results but did not systematically explore data augmentation, input resolution, class-weight configurations in BCE loss, or classification thresholds. Although multiple architectures were tested, including EfficientNetV2 (Small, Large), ResNet (18, 50), DenseNet (121, 201), and SwinV2 (Tiny, Small), scaling within the EfficientNet family was only partially examined. In this work, we focus on EfficientNetV2 backbones, systematically analyzing architectural scaling alongside training configurations. This addresses prior limitations and establishes a more comprehensive and reproducible performance benchmark.

C. Explainable Artificial Intelligence (XAI)

Explainable Artificial Intelligence (XAI) seeks to make AI models more transparent and interpretable, a critical requirement in medical imaging for clinical trust, accountability, and diagnostic support. Approaches can be based on *hand-crafted features* (e.g., radiomics descriptors [26]) or *intrinsically interpretable models* [27] such as decision trees. Deep neural networks, however, typically require post hoc methods to generate human-understandable explanations. These vary by *output modality*-ranging from text-based rationales [28] to visual explanations such as heatmaps and bounding boxes [22]-and by design, being either *model-specific* (e.g., Grad-CAM [29]) or *model-agnostic* [3]. Post hoc model-agnostic methods like LIME [30] and SHAP [31] are widely used, but in medical imaging, CAM-based visualizations remain the most common.

For brain hemorrhage, prior XAI studies have relied almost exclusively on CAM methods to highlight regions influencing classification, typically using datasets such as RSNA or CQ500. These explanations were mainly assessed qualitatively by radiologists, since pixel-level ground truth was lacking [32]. Extensions such as probabilistic CAMs or weakly supervised localization [11] improved visualization but remained limited by small-scale validation.

The Hemorica dataset, which uniquely provides both slice-level labels and pixel-level segmentation masks, enables the first systematic *quantitative* evaluation of CAM methods for hemorrhage. CAM outputs can now be benchmarked against ground truth using overlap metrics such as Dice and IoU, moving beyond subjective inspection toward reproducible evaluation. This study therefore focuses on *visual, model-agnostic* CAM techniques, leveraging Hemorica's detailed annotations to provide a comprehensive comparison of localization performance and address a key gap in prior literature.

III. METHODOLOGY

A. Dataset

The Hemorica dataset comprises 327 non-contrast head CT examinations collected from multiple institutions, accompanied by both slice-level labels and pixel-level segmentation masks. In total, it contains 12,067 axial slices, of which 2,679 are labeled as hemorrhage-positive and 9,388 as hemorrhage-negative. Positive slices are further categorized

into five hemorrhage subtypes: intracerebral hemorrhage (ICH, 1,616 slices), intraventricular hemorrhage (IVH, 471 slices), epidural hemorrhage (EPH, 157 slices), subdural hemorrhage (SDH, 752 slices), and subarachnoid hemorrhage (SAH, 253 slices). However, due to the substantial class imbalance among subtypes and the relatively low frequency of some categories, we adopt a binary classification scheme (hemorrhage vs. non-hemorrhage) to maximize data availability for the positive class and improve model robustness. The dataset presents several challenges, including a pronounced class imbalance (with negatives comprising over 75% of slices), variability in hemorrhage size and shape, and differences in CT acquisition parameters across scanners and institutions. These factors make Hemorica a realistic yet challenging benchmark for developing clinically relevant detection models.

For model development, the dataset was partitioned at the *patient level* to avoid data leakage across scans from the same subject. A stratified split ensured that hemorrhage-positive and negative cases were proportionally represented in each set. Specifically, 80% of the studies (261 patients, 9,693 slices) were used for training and 20% of the studies (66 patients, 2,374 slices) were reserved for testing, with no overlap of patient scans between sets. This strategy preserves independence between subsets while maintaining adequate representation of hemorrhage subtypes in both training and testing.

B. Class Activation Mapping (CAM) Methods

Class Activation Mapping (CAM) techniques are widely used for interpreting deep neural networks, particularly in medical imaging where explainability is critical. These methods highlight discriminative image regions that drive a model's decision by combining layer activations with gradients, perturbations, or statistical decompositions. The resulting saliency maps provide insight into spatial features influencing the classifier's output, enabling both qualitative and quantitative evaluation of model trustworthiness [8].

We benchmarked the following nine representative CAM variants:

- **GradCAM** [29]: Weights feature maps by the average gradient of the class score with respect to activations, producing class-discriminative localization maps.
- **HiResCAM** [33]: Forms maps by taking the element-wise product of activations and backpropagated gradients, preserving spatial correspondence across all locations before summation.
- **GradCAM-ElementWise** [34]: Extends GradCAM by applying the element-wise product of activations and gradients, followed by ReLU and channel summation.
- **GradCAM++** [35]: Improves GradCAM by incorporating higher-order derivatives, allowing distinct regions within the same class to contribute independently to the map.
- **XGradCAM** [36]: Enforces axiomatic properties by combining normalized activations with gradients, ensuring proportional contributions to the localization map.

- **AblationCAM** [37]: Estimates channel importance via iterative ablation and measuring class score drops, with a batched implementation for efficiency.
- **EigenCAM** [38]: Uses the first principal component of the activation space to generate class-agnostic saliency maps.
- **EigenGradCAM** [34]: Multiplies EigenCAM's principal component with gradient signals to obtain cleaner class-specific maps.
- **LayerCAM** [39]: Assigns pixel-level weights by multiplying activations with their corresponding positive gradients, enabling finer localization in early or intermediate layers.

Each of these CAM variants emphasizes different aspects of localization accuracy and spatial resolution. By applying them to the last three stages of the network, it becomes possible to analyze how explanation quality varies across methods and stages, and to determine which approaches provide the most reliable localization of hemorrhage regions.

C. EfficientNet

EfficientNet is a family of convolutional neural networks that introduced compound scaling to jointly balance depth, width, and input resolution in a principled manner [40], achieving high accuracy with strong computational efficiency. This study employs EfficientNetV2 [41], which improves training speed, parameter utilization, and performance on high-resolution inputs through fused-MBConv layers in early stages and MBConv layers in deeper stages. Six variants are evaluated-B0, B1, B2, B3, S, and L-ranging from lightweight models suitable for constrained settings to larger architectures with higher representational capacity. Their comparison under uniform training and evaluation protocols enables systematic assessment of the trade-offs between efficiency and performance for brain hemorrhage detection.

D. Model Architecture

For the classification backbone, EfficientNetV2-S was employed due to its strong balance between accuracy and computational efficiency. Figure 1 illustrates the architecture, showing the progression of feature map dimensions (height $\times$ width) from the input resolution of 512×512 down to the final stage. Each stage is composed of repeated building blocks, where N denotes the number of repetitions of that block (e.g., MBConv or FusedMBConv) within the stage. As shown in Fig. 4, EfficientNetV2-S achieved the highest area under the precision-recall curve (AUC) among all EfficientNetV2 variants, confirming its superiority as the final chosen model for our experiments. From a theoretical perspective, deeper stages in CNN-based architectures encode higher-level semantic features that are more closely aligned with class-discriminative regions, while earlier stages primarily capture low-level textures and edges. Consequently, the final three convolutional stages were selected as target stages for CAM-based experiments, since they provide the most meaningful and interpretable activations for localization tasks.

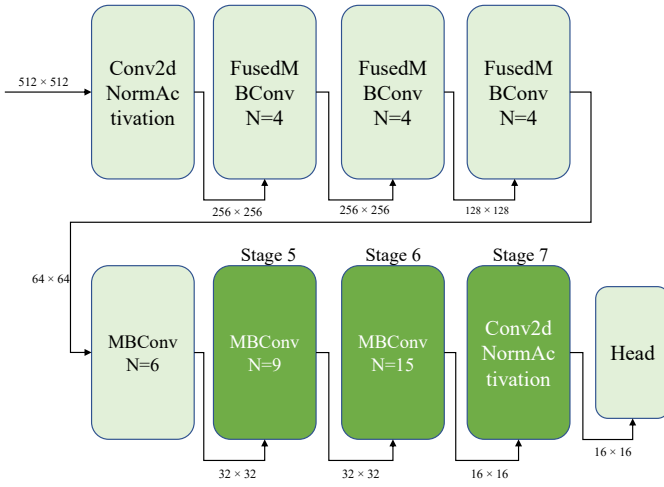


Fig. 1: EfficientNetV2-S architecture used in this study. The diagram shows input and output feature map dimensions at each stage. N indicates the number of repetitions of the given block type. The last three stages (5, 6, and 7) correspond to the stages used for CAM-based analysis.

E. Metrics

The quality of Class Activation Maps (CAMs) was evaluated using pixel-level, bounding-box, and detection metrics. For each slice, CAM heatmaps were thresholded at τ to produce binary masks M , compared against ground-truth masks G from Hemorica.

1) Pixel-wise Metrics:

$$\text{IoU} = \frac{|M \cap G|}{|M \cup G|} \quad (1) \quad , \quad \text{Dice} = \frac{2|M \cap G|}{|M| + |G|} \quad (2)$$

2) *Bounding-box Metrics*: To evaluate coarse localization, binary masks were converted into tight bounding boxes around connected regions. The overlap between predicted and ground-truth boxes was then quantified using IoU and Dice at the bounding-box level.

3) *Loose Hit Rate*: Loose Hit Rate measures whether a CAM mask overlaps the lesion at all. It is assigned a value of 1 if any intersection with the ground truth exists, and 0 otherwise, and is reported as the average across slices.

4) *Top-5 F1 Score*: To assess the stability of training, we also report the *Top-5 F1 Score*. Specifically, after training each model, the F1 scores from all epochs are ranked, and the five epochs with the highest F1 values are selected. The final Top-5 F1 is reported as the mean μ and standard deviation σ across these epochs:

$$\text{Top-5 F1} = \mu \pm \sigma$$

This provides a robust estimate of peak performance while mitigating sensitivity to a single best epoch.

5) *Top-5 Precision*: To complement the F1 score, we also report the *Top-5 Precision*. After training, all epoch-wise Precision values are ranked, and the five highest are selected.

The final Top-5 Precision is expressed as the mean μ and standard deviation σ across these epochs:

$$\text{Top-5 Precision} = \mu \pm \sigma$$

6) *Top-5 Recall*: Similarly, we report the *Top-5 Recall*. Epoch-wise Recall values are ranked, and the five highest are chosen. The final Top-5 Recall is then given as the mean μ and standard deviation σ across these epochs:

$$\text{Top-5 Recall} = \mu \pm \sigma$$

IV. EXPERIMENTS

Various factors including model architecture (EfficientNetV2-B0, B1, B2, B3, S, and L), input image resolution, and class imbalance can affect model performance. In this study, experiments were designed to isolate and quantify the influence of these factors on classification results.

All CT scans were preprocessed before training. Each scan was windowed to a brain window with center 40 and width 80 to enhance soft tissue contrast. Images were normalized to the range $[0, 1]$ and standardized to zero mean and unit variance. Slices were resized to the target input resolution (224, 448, or 512 pixels) depending on the experiment. Ground-truth segmentation masks were resized with nearest-neighbor interpolation to preserve categorical labels. No histogram equalization or color augmentation was applied, since CT images are grayscale and their intensity values are diagnostically meaningful.

Before describing the experimental variations, a baseline configuration was defined and held constant across all experiments unless explicitly changed. The backbone network was EfficientNetV2-B0, trained with a batch size of 16, an initial learning rate of 0.001, and the Adam optimizer for 40 epochs. No data augmentation was applied. The loss function employed a positive class weight of 1.0, and the input image resolution was set to 224×224 . A fixed classification threshold of 0.5 was used to determine positive predictions.

On top of this baseline, several factors were systematically varied to assess their impact on classification performance. Models were trained with input resolutions of 224×224 , 448×448 , and 512×512 pixels to evaluate how higher spatial detail influences predictive accuracy. To investigate the role of model capacity, multiple EfficientNetV2 variants (B0, B1, B2, B3, S, and L) were compared, providing insight into the trade-off between performance and computational cost. Class imbalance was addressed by altering the positive weight parameter of the binary cross-entropy loss function, thereby emphasizing hemorrhage-positive slices during training and allowing the effect on the sensitivity–specificity balance to be observed. Classification probabilities were binarized at thresholds of 0.3, 0.5, and 0.7 to determine the operating point that maximizes the F1-score. Finally, data augmentation was explored through horizontal flipping ($p = 0.5$), rotation (limit 15°), and shift-scale-rotate transformations (shift limit 0.1, scale limit 0.1, rotation limit 15°). Color jittering

was deliberately excluded, as the most distinctive feature for hemorrhage detection in CT images is intensity (grayscale value), and altering it could obscure critical diagnostic cues. These experiments collectively quantified how augmentation contributes to generalization and robustness.

Each experimental variation was compared against the baseline configuration: EfficientNetV2-B2 without augmentation, positive weight = 1, input resolution 224×224 , and classification threshold 0.5. Sequentially integrating the proposed factors increased the F1-score from 0.8924 to 0.9258, an absolute improvement of 0.0334.

In the next stage, the reasoning process of the optimized classifier was investigated. Nine widely used Class Activation Mapping (CAM) methods were applied to the last three stages of the model. For each CAM method and target stage, we scanned 50 linearly spaced candidate thresholds $\tau \in [0.1, 0.9]$ and, at each τ , binarized the heatmaps and computed a pixel-wise Dice score against the annotations. The optimal threshold was then defined as $\tau^* = \arg \max_{\tau} \text{Dice}(\tau)$. When a training split was available, τ^* was estimated *on training data only* and then fixed for the final evaluation on the test set. These CAM-derived masks were then evaluated against Hemorica ground truth using pixel-wise, bounding-box, and loose-hit metrics, thereby quantifying how well each interpretability technique aligns with hemorrhage regions.

TABLE I: Optimal thresholds for CAM binarization across methods and stages.

CAM Method	stage 7	stage 6	stage 5
GradCAM	0.49	0.57	0.61
HiResCAM	0.49	0.38	0.21
GradCAM-ElementWise	0.52	0.39	0.26
GradCAM++	0.49	0.49	0.56
XGradCAM	0.52	0.57	0.64
AblationCAM	0.48	0.30	0.48
EigenCAM	0.64	0.51	0.70
EigenGradCAM	0.52	0.72	0.31
LayerCAM	0.51	0.38	0.25

V. RESULTS

This section reports the experimental findings in two parts: (i) classification model performance under different training configurations and EfficientNetV2 backbones, and (ii) quantitative evaluation of CAM-based localization methods.

A. Classification Performance

Table II summarizes the unified ablation study. The baseline configuration, using input size 224×224 , no augmentation, and BCE positive weight of 1, achieved a mean top-5 F1 score of 0.8924. Increasing the input resolution to 448×448 and 512×512 consistently improved performance, yielding gains of +0.0135 and +0.0166, respectively. Similarly, introducing data augmentation (random flips, rotations, and scaling) boosted the F1 score to 0.9106, the highest among single-factor ablations. In contrast, modifying the BCE positive weight to 2 or 3 slightly degraded performance, suggesting

that the baseline loss weighting already provided a balanced treatment of the hemorrhage and non-hemorrhage classes.

Figure 3 compares different EfficientNetV2 architectures under the baseline configuration. EfficientNetV2-S achieved the best performance with a mean top-5 F1 score of 0.9090, outperforming smaller (B0-B3) and larger (L) variants. This indicates that the S variant offers the most favorable trade-off between representational capacity and overfitting for this dataset.

Figure 4 shows the precision-recall (PR) curves across classification thresholds (0.3-0.7). All models demonstrated robustness to threshold variation, with minimal sensitivity to changes in decision boundary. Importantly, EfficientNetV2-S trained at 512×512 with augmentation achieved the highest area under the PR curve (AUC), confirming it as the final selected model for downstream localization experiments.

B. CAM-based Localization

Tables IV and III report the performance of nine CAM variants across three selected convolutional layers of EfficientNetV2-S. Metrics were computed both globally (aggregated across all slices) and on a per-slice basis. Pixel-wise metrics (IoU and Dice) measure the exact spatial overlap between CAM-derived masks and ground-truth segmentations, providing the most fine-grained evaluation of localization accuracy. Bounding-box metrics offer a coarser assessment by quantifying whether the predicted lesion extent falls within the correct anatomical region, facilitating comparison with detection-based methods. Finally, the Loose Hit Rate provides a binary indicator of whether any overlap exists between predicted and ground-truth regions, reflecting sensitivity at the lesion level. Together, these complementary metrics capture different dimensions of localization performance, from precise boundary alignment to coarse region detection.

The strongest results across all evaluation metrics were consistently obtained from stage five of the EfficientNetV2-S architecture (see Fig. 1). At this depth, HiResCAM achieved the highest bounding-box overlap, while AblationCAM produced the best pixel-level Dice and IoU scores. These findings indicate that intermediate stages, rather than the final stage, provide the most informative representations for localization.

The Loose Hit Rate was consistently high (> 0.95 for most methods), demonstrating that hemorrhage regions were reliably detected across slices. Importantly, the best-performing CAMs achieved pixel Dice scores around 0.57 and IoU scores around 0.40, which represent strong localization quality given that the models were trained solely for classification without segmentation supervision. When compared with transformer-based weakly supervised localization approaches, which typically report Dice scores of approximately 0.455 ± 0.019 on brain hemorrhage datasets [11], our best CAM-based methods demonstrate competitive or superior overlap performance despite relying only on convolutional backbones.

In summary, the most accurate localization was observed at stage five of EfficientNetV2-S, with HiResCAM and AblationCAM providing the top-performing results depending on the

TABLE II: Unified ablation study (mean $\pm$ std of top-5 **F1 Score**, Precision, and Recall). Row 1 shows the **baseline model** (224×224 , no augmentation, BCE weight = 1). Subsequent rows vary one factor at a time: input resolution (Rows 2–3), data augmentation (Row 4), and BCE positive weight (Rows 5–6). The last columns (Δ F1, Δ Precision, Δ Recall) indicate improvement (green) or degradation (red) relative to the baseline.

Experiment	Input Res.	Aug.	Pos. W.	F1 Score (Top-5)	Δ F1	Precision (Top-5)	Δ Precision	Recall (Top-5)	Δ Recall
Baseline	224×224	–	1	0.8924 ± 0.0033	–	0.9666 ± 0.0093	–	0.8876 ± 0.0108	–
Input Resolution	448×448	–	1	0.9059 ± 0.0018	+0.0135	0.9662 ± 0.0067	-0.0004	0.9147 ± 0.0095	+0.0271
	512×512	–	1	0.9090 ± 0.0057	+0.0166	0.9644 ± 0.0079	-0.0022	0.9075 ± 0.0043	+0.0199
Augmentation	224×224	✓	1	0.9106 ± 0.0022	+0.0182	0.9769 ± 0.0024	+0.0103	0.9147 ± 0.0162	+0.0271
Positive Class Weight	224×224	–	2	0.8910 ± 0.0018	-0.0014	0.9563 ± 0.0037	-0.0103	0.9151 ± 0.0113	+0.0275
	224×224	–	3	0.8870 ± 0.0013	-0.0054	0.9392 ± 0.0081	-0.0274	0.9147 ± 0.0133	+0.0271

TABLE III: Comparison of nine CAM methods across the last three stages (Stage 7, Stage 6, Stage 5) of EfficientNetV2-S using localization metrics computed **globally** across the dataset. Pixel-level metrics (Dice, IoU) measure fine-grained overlap, bounding-box metrics assess coarse alignment, and Loose Hit Rate reflects whether any hemorrhage region was detected. Best results for each metric are shown in **bold**.

CAM Method	Loose Hit Rate			Pixel Dice			Pixel IoU			BBBox Dice			BBBox IoU		
Stage	[7]	[6]	[5]	[7]	[6]	[5]	[7]	[6]	[5]	[7]	[6]	[5]	[7]	[6]	[5]
GradCAM	0.9583	0.9279	0.4573	0.3640	0.3119	0.1739	0.2225	0.1848	0.0953	0.4117	0.4058	0.2154	0.2592	0.2546	0.1207
HiResCAM	0.9602	0.9620	0.9753	0.4219	0.4422	0.5219	0.2673	0.2839	0.3530	0.4883	0.5176	0.5723	0.3230	0.3492	0.4009
GradCAMElementWise	0.9620	0.9658	0.9734	0.4508	0.4523	0.5117	0.2910	0.2922	0.3438	0.5049	0.5248	0.5568	0.3377	0.3557	0.3858
GradCAM++	0.9564	0.9279	0.3814	0.3650	0.3066	0.0794	0.2232	0.1811	0.0413	0.4135	0.4206	0.1175	0.2606	0.2663	0.0624
XGradCAM	0.7173	0.6357	0.5750	0.0772	0.2202	0.2215	0.0401	0.1238	0.1246	0.1303	0.2736	0.2421	0.0697	0.1585	0.1377
AblationCAM	0.9051	0.8994	0.8956	0.3231	0.1065	0.5744	0.1927	0.0562	0.4029	0.3672	0.2048	0.5679	0.2249	0.1141	0.3966
EigenCAM	0.8482	0.9507	0.6528	0.2244	0.4752	0.2311	0.1264	0.3116	0.1307	0.2197	0.5251	0.2321	0.1234	0.3561	0.1313
EigenGradCAM	0.9469	0.1822	0.8065	0.4247	0.0117	0.2258	0.2696	0.0059	0.1272	0.4770	0.0357	0.2879	0.3132	0.0182	0.1681
LayerCAM	0.9602	0.9696	0.9753	0.4312	0.4398	0.4909	0.2749	0.2818	0.3253	0.4919	0.5277	0.5384	0.3261	0.3584	0.3684

metric. Complete quantitative details are provided in Tables IV and III.

VI. CONCLUSION

In this work, a comprehensive study on brain hemorrhage classification and localization was conducted using EfficientNetV2 backbones combined with Class Activation Mapping (CAM) techniques. Through an extensive set of controlled ablation experiments, input resolution and data augmentation were shown to be the most influential factors in improving classification performance, while increasing the BCE positive class weight did not yield further benefits. Among the tested

models, EfficientNetV2-S trained at 512×512 resolution with augmentation achieved the best F1 score and the most stable precision–recall behavior across thresholds.

Beyond classification, a systematic comparison of CAM variants was performed across multiple stages and evaluation metrics, both globally and averaged per slice. This analysis showed that HiResCAM and AblationCAM achieved the strongest localization results in deeper stages of the network. By quantifying localization with pixel-wise and bounding-box metrics, a rigorous benchmark for explainability methods in the context of brain hemorrhage was established.

Overall, the results indicate that careful architectural choices

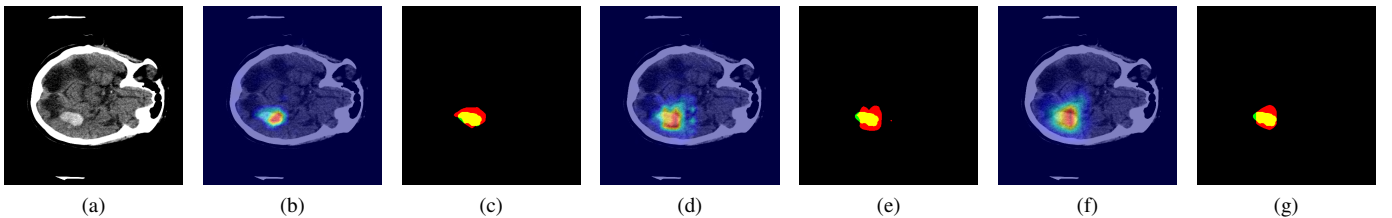


Fig. 2: Example of Class Activation Mapping (CAM) visualizations for a hemorrhage-positive CT slice using HiResCAM across three different network depths. (a) Original CT slice; (b, c) overlay and binary mask from stage 5; (d, e) overlay and mask from stage 6; (f, g) overlay and mask from stage 7. In overlays, the red regions correspond to CAM-predicted hemorrhage, the green regions represent the ground-truth segmentation mask, and the yellow regions denote the intersection between prediction and ground truth. This progression illustrates how CAM maps evolve across successive convolutional stages, with earlier stages often capturing broader regions and deeper stages providing more focused localization.

TABLE IV: Comparison of nine CAM methods across the last three stages (Stage 7, Stage 6, Stage 5) of EfficientNetV2-S using multiple localization metrics **averaged per slice**. Pixel-level metrics (Dice, IoU) assess mask overlap, while bounding-box metrics assess coarse localization. The best result for each metric is highlighted in **bold**.

CAM Method	Pixel Dice			Pixel IoU			BBox Dice			BBox IoU		
Stage	[7]	[6]	[5]	[7]	[6]	[5]	[7]	[6]	[5]	[7]	[6]	[5]
GradCAM	0.3815	0.3217	0.1574	0.2581	0.2088	0.1082	0.4117	0.4058	0.2154	0.2592	0.2546	0.1207
HiResCAM	0.3838	0.4119	0.4701	0.2594	0.2801	0.3326	0.4883	0.5176	0.5723	0.3230	0.3492	0.4009
GradCAMElementWise	0.4012	0.4172	0.4682	0.2750	0.2842	0.3303	0.5049	0.5248	0.5568	0.3377	0.3557	0.3858
GradCAM++	0.3775	0.3495	0.0898	0.2547	0.2308	0.0594	0.4135	0.4206	0.1175	0.2606	0.2663	0.0624
XGradCAM	0.2789	0.2258	0.1875	0.1901	0.1548	0.1291	0.1303	0.2736	0.2421	0.0697	0.1585	0.1377
AblationCAM	0.3529	0.0902	0.5043	0.2364	0.0488	0.3759	0.3672	0.2048	0.5679	0.2249	0.1141	0.3966
EigenCAM	0.3537	0.4537	0.3103	0.2441	0.3219	0.2266	0.2197	0.5251	0.2321	0.1234	0.3561	0.1313
EigenGradCAM	0.4069	0.0156	0.2907	0.2812	0.0092	0.2019	0.4770	0.0357	0.2879	0.3132	0.0182	0.1681
LayerCAM	0.3904	0.4083	0.4595	0.2655	0.2763	0.3226	0.4919	0.5277	0.5384	0.3261	0.3584	0.3684

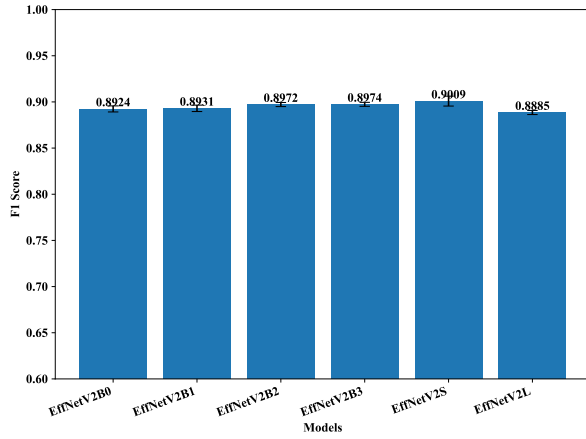


Fig. 3: Model comparison (mean $\pm$ std of top-5 **F1 Score**). This bar chart compares the performance of different EfficientNetV2 variants under the baseline configuration (input resolution 224×224 , no augmentation, and BCE positive weight of 1), where the only factor changed is the model architecture. Among all evaluated backbones, **EfficientNetV2-S** achieves the highest F1 score, indicating superior classification performance for this task.

and training configurations, combined with interpretable CAM-based visual explanations, can lead to improved classification accuracy and clinically meaningful model interpretability. Further improvements may be obtained through post-processing techniques designed to suppress spurious activations and refine localization quality.

Finally, a failure analysis revealed two common error modes. First, some false positives occurred when the CT machine's scan lines or artifacts were mistaken for hemorrhages. Second, certain hemorrhages with pixel intensities similar to surrounding brain tissue were missed, leading to false negatives. These cases highlight the challenges of differentiating subtle imaging artifacts from true pathology, and suggest that integrating artifact suppression or intensity-sensitive preprocessing could further improve robustness.

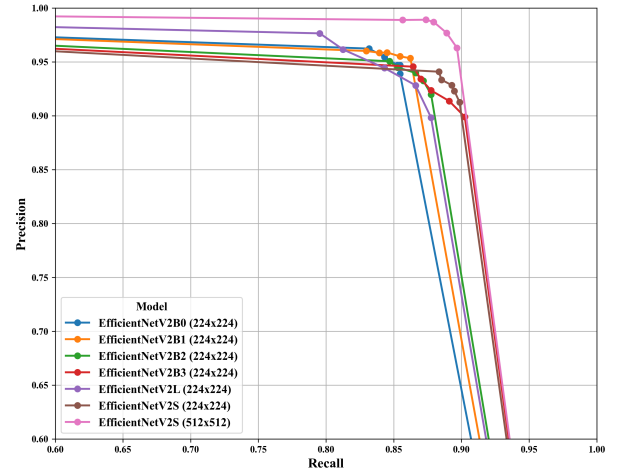


Fig. 4: Precision-Recall curves of EfficientNetV2 variants at the epoch corresponding to the best F1 score. Each curve shows the trade-off between precision and recall as the classification threshold is varied from 0.3 (rightmost point) to 0.7 (leftmost point). The results demonstrate that all models are relatively insensitive to small changes in threshold, maintaining high precision across a wide recall range. Among the compared architectures, **EfficientNetV2-S** (512×512) (pink curve) achieves the highest area under the precision-recall curve (AUC), confirming its superiority as the final chosen model.

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